Reference Card Pack

Decision matrices and quick references for all six sessions

Azure Storage, Data Protection, and Resilience at Scale: University of Minnesota custom course

Card	Session	Use it when
1. Azure Storage decision matrix	1	Choosing account type, namespace, tier, and redundancy for a data class
2. Irreversible decisions and scale targets	1	Reviewing a design before data lands; sharding beyond one account
3. Access tiers and lifecycle quick reference	1 and 6	Writing or reviewing a lifecycle rule; estimating early deletion and rehydration
4. Data movement decision tree	2	Picking the tool for a migration by volume, timeline, and network
5. Immutability and identity quick reference	2	Configuring WORM, ABAC, and SAS governance
6. Protection pattern reference	3	Which vault, which control, which workload
7. Ransomware resilience controls	3	Explaining the control stack and what each refuses
8. Redundancy option comparison	4	Choosing LRS, ZRS, GRS, GZRS, read-access variants, object replication
9. RPO and RTO design method	4	Building or reviewing the resilience matrix
10. Oracle on Azure decision guide	5	Choosing the Oracle path and the backup pattern
11. FinOps and governance quick reference	6	Cost levers, reserved capacity rules, policy effects, pipeline pattern
12. Azure CLI cheat sheet	All	The commands used in the demos and labs
13. Certification pathway	6	Mapping roles to AZ-104, AZ-305, DP-300, and SC-500

Every figure on these cards reflects Microsoft Learn as of October 2026. Limits, regional availability, and preview status change; re-verify before quoting a number to a stakeholder. Prices are never stated on these cards; use the cost model workbook with current pricing.

Card 1. Azure Storage decision matrix

Session 1: Azure Storage Architecture for Multi-Petabyte Portfolios

Start every design from the account type, then namespace, then default tier, then redundancy. Each row is a question; each column is the answer that fits.

Decision	Choose this	When	Watch out
Account type	Standard general-purpose v2	Blobs, files, queues, tables; all four access tiers plus smart tier	GPv1 and legacy Blob storage accounts reached full retirement in October 2026; GPv2 transaction pricing differs
	Premium block blob	Small objects, high transaction rates, low latency (analytics scratch, media)	No access tiers; LRS and ZRS only
	Premium page blob	Random-write VHD-style workloads	Most VM disks use managed disks
	SSD (premium) file shares, FileStorage kind	NFS shares; latency-sensitive SMB	Standalone Microsoft.FileShares resource exists for SSD NFS shares
Namespace	Flat	Object workloads, backup targets, archives; every Blob data-protection feature available	None
	Hierarchical (Data Lake Storage Gen2)	Analytics on Spark, Databricks, Synapse; directory renames; POSIX ACLs; NFS 3.0 and SFTP	One-way; no versioning, change feed, object replication, point-in-time restore, version-level WORM; tags and snapshots preview; protection is vaulted backup for Data Lake Storage
Default tier	Hot, Cool, or Cold at account level; Archive per blob	Hot for active, Cool for completed projects, Cold for reference read a few times a year	Archive is offline, hours to rehydrate, 180-day minimum
Redundancy	ZRS	Zone loss tolerance, active data, backup targets	Premium supports LRS and ZRS only; redundancy is account-wide
	GRS or GZRS	Regional loss tolerance for irreplaceable data	Asynchronous; recent writes lost on unplanned failover; secondary is the paired region
	RA-GRS or RA-GZRS	Read access during a primary outage	Application must handle the secondary endpoint
File services	Azure Files SMB or NFS	Departmental shares; hybrid with File Sync	Provisioned v2 billing recommended for new shares; NFS needs SSD; one identity source per account
	Azure NetApp Files	Tier-1 NAS, databases, SAP, engineering apps	Capacity pool then volumes; Standard, Premium, Ultra, Flexible service levels
	Azure Managed Lustre	HPC parallel scratch	Scratch only; Blob holds the durable copy

Security baseline for every account

HTTPS only; minimum TLS 1.2 (TLS 1.0 and 1.1 are no longer accepted); anonymous blob access off; shared key access off where tooling allows; private endpoints with firewall default deny; private DNS zones planned; Microsoft-managed or customer-managed keys.

Card 2. Irreversible decisions and scale targets

Session 1

Treat these as architecture decisions, not configuration settings.

Decision	Reversible?	Consequence
Account kind and performance tier	No	Moving standard to premium means migrating to a new account
Hierarchical namespace	One-way	Cannot be disabled; loses Blob-native protection features
Redundancy	Changeable, but account-wide	Mixed durability needs force separate accounts; some conversions take time and block failover meanwhile
Account name	No	Appears in every endpoint URL; 3 to 24 lowercase letters and digits, globally unique
Locked immutability policy	No	Blob and account cannot be deleted until expiry
Smart tier, auto-protection, AlwaysOn soft delete	No or restricted	Each is a one-way switch in its service

Scale targets, standard GPv2 (defaults; higher by request; verify for your region)

Target	Value
Storage accounts per region per subscription	250 by default, 500 by request
Maximum account capacity	5 PiB by default
Request rate per account	40,000 per second in major regions (Central US, North Central US, East US, East US 2, West US 2, West US 3 among them); 20,000 elsewhere
Ingress per account	60 Gbps in those regions; 25 Gbps elsewhere
Egress per account	200 Gbps in those regions; 50 Gbps elsewhere
Private endpoints per account	200
Single block blob	About 190.7 TiB; target 3,000 requests per second per blob

At a partition limit you see 503 Server Busy or 500 Operation Timeout: retry with exponential backoff and avoid sequential names that create hot partitions.

Sharding patterns

Pattern	Shard by	Why
Data domain	Research unit	Cost, access, and redundancy map cleanly to owners
Protection class	Redundancy and WORM presence	Redundancy is account-wide; last access tracking cannot coexist with WORM
Request profile	Analytics versus archive	Isolates request rates
Catalog and template	Always	Map datasets to accounts; deploy every account from the same module

Card 3. Access tiers and lifecycle quick reference

Sessions 1 and 6

Tier	Online	Minimum retention	Typical data	Notes
Hot	Yes	None	Active project data	Highest storage, lowest access cost
Cool	Yes	30 days	Completed projects still consulted	Early deletion prorated
Cold	Yes	90 days	Reference data read a few times a year	Fast retrieval; not eligible for reserved capacity
Archive	No	180 days	Retained-for-record, decades of telemetry	Standard rehydration up to 15 hours; High may be under one hour for objects under 10 GB; raise Standard to High while pending
Smart tier	Yes	None within Hot, Cool, Cold	Unpredictable access	GPv2 with ZRS, GZRS, or RA-GZRS; Hot to Cool after 30 days, Cold after 90, back to Hot on access; never Archive; ignores lifecycle tiering rules; Hot access rates plus monitoring meter; excluded from reserved capacity

Lifecycle rule anatomy

Part	Options
Filters	Blob types (block, append); prefix match; blob index tag match (preview on HNS accounts)
Conditions	Days since modification, creation, or last access (last access requires tracking, which is incompatible with immutable storage)
Actions	Tier to Cool, Cold, or Archive; delete; for base blobs, versions, and snapshots
Execution	Policy read and written in full; runs once per day; first run after creation may take up to 24 hours

Model the money before enabling a rule

Step	What to do
1	Inventory capacity, object counts, and last access by dataset (blob inventory reports at scale)
2	Classify datasets with tags (research unit, retention class)
3	Price steady-state capacity per tier with current pricing
4	Add one-time tier-change transactions and early deletion exposure
5	Review the affected datasets by name, then commit

Surprise line items

Early deletion (leaving a tier before its minimum); rehydration priority; millions of small objects (transactions and tier-change costs); hidden capacity in versions, snapshots, and soft-deleted data; auto-tiering back to Hot inside the minimum period; egress for copies out of Azure.

Card 4. Data movement decision tree

Session 2

Microsoft's selection guidance weighs data size, available bandwidth, and whether the transfer is one-time or recurring.

Question	If yes	If no
Can the network move the volume within the timeline at the bandwidth you actually have? (10 TB per day per 1 Gbps at full utilization)	Online tools	Data Box family
Is the source an NFS or SMB share needing repeated syncs and cutover?	Azure Storage Mover	AzCopy for scripted copies and sync
Is it a recurring feed with transformation or orchestration?	Azure Data Factory (self-hosted integration runtime for on-premises)	AzCopy or Storage Mover
Is the source a Windows file server moving to Azure Files with ongoing caching?	Azure File Sync (no downtime)	Storage Mover or AzCopy
Is the volume petabytes with limited bandwidth or a deadline?	Data Box 525 (several) or Data Box 120	Online tools

Tool	Best for	Notes
AzCopy	One-off copies, scripted jobs, incremental sync; Blob, Files, between accounts	Tune concurrency and buffers; Microsoft Entra ID auth
Azure Storage Mover	NFS and SMB share migration, TB to PB, over the network	Cloud service plus on-premises agent VM; one resource orchestrates many agents
Azure Data Factory	Orchestrated and recurring transfers with transforms	Self-hosted integration runtime reaches on-premises stores; monitoring and retry built in
Azure File Sync	Windows file servers to Azure Files with cloud tiering	Microsoft recommends HDD provisioned v2 shares for File Sync
Data Box Disk	Smaller offline volumes	SSDs shipped to you
Data Box 120	120 TB usable, interfaces up to 100 Gbps	Next-generation device
Data Box 525	525 TB usable	Order several for petabyte moves
Retired	Data Box Heavy; original 80 TB Data Box retiring region by region	Existing orders remain supported; check regional availability

Connectivity and egress

Option	Use for	Egress
ExpressRoute private peering	Virtual networks and private endpoints; sustained storage traffic	Metered or unlimited plans price outbound data differently
ExpressRoute Microsoft peering	Public Microsoft service endpoints	Same
VPN Gateway	Encrypted tunnels over the internet; throughput limited by SKU	Internet egress rates apply
Any	Data into Azure is generally free; out of a region is billed; cross-region replication and reads carry transfer charges	Model before designs that routinely pull data back

Card 5. Immutability and identity quick reference

Session 2

Policy	Scope	Interval	Supported accounts	Notes
Time-based retention, container-level	Container	1 to 146,000 days (400 years)	GPv2, premium block blob, HNS	Up to 10,000 containers per account; immutable blobs on HNS cannot be renamed or moved
Time-based retention, version-level	Account, container, or version	Per version	GPv2, premium block blob	Requires versioning; not on HNS
Legal hold	Container	Until cleared	Same as container-level	Can apply with a retention policy
Any			Not on NFS 3.0 or SFTP accounts	Not with last access tracking or point-in-time restore

State	Can change interval	Can delete policy	Account deletable	Guidance
Unlocked	Yes	Yes	Yes (blobs still protected)	Test here; lock within about 24 hours
Locked	Extend up to five times, never shorten	No	No, while any locked policy exists	Up to 35 seconds to take effect after creation

Identity for storage

Need	Mechanism	Notes
Azure Files SMB with existing identities	On-premises AD DS, Microsoft Entra Domain Services, or Microsoft Entra Kerberos (hybrid and cloud-only)	One source per account; Kerberos AES-256 recommended; not for NFS; Entra Kerberos ACL changes need domain controller line of sight
Share and file permissions	Share-level Azure RBAC plus directory and file Windows ACLs	Entra-only groups work only with synced users
Blob access for people	Storage Blob Data Reader, Contributor, Owner at account, container, or resource group scope, to groups	Data plane is separate from management-plane Contributor
Precise access without ACL sprawl	ABAC conditions on blob index tags or container names	Example: read only where ResearchUnit tag equals PGRL; exclude Blob.List so listing works
Automation	Managed identities (system or user assigned)	No keys or connection strings in scripts
Delegated access	User delegation SAS (Entra-signed)	Short expiry; stored access policies for service SAS; disable shared key access to block key-signed SAS

ABAC condition for tag-scoped reads (from Microsoft Learn):

```
((!(ActionMatches{'Microsoft.Storage/storageAccounts/blobServices/containers/blobs/read'} AND NOT SubOperationMatches{'Blob.List'})) OR (@Resource[Microsoft.Storage/storageAccounts/blobServices/containers/blobs/tags:ResearchUnit<$key_case_sensitive$>] StringEquals 'PGRL'))
```

Card 6. Protection pattern reference: which vault, which control, which workload

Session 3

Create the vault that matches the datasource. A vault protects resources in its own region and, for most workloads, its own subscription.

Workload	Vault or mechanism	Backup tier	Key controls
On-premises files, folders, volumes, system state	Recovery Services vault via MARS agent	Vaulted	Immutable vault, enhanced soft delete, MUA, passphrase custody, offline seeding
On-premises applications (SQL Server, Exchange, SharePoint, Hyper-V, VMware)	MABS to Recovery Services vault	Local disk tier plus vault	Immutable vault, MUA, bandwidth throttling
Azure VMs	Recovery Services vault	Vaulted (instant restore snapshots plus vault)	Immutable vault, enhanced soft delete, MUA, cross-region restore with GRS
SQL Server, SAP HANA, SAP ASE in Azure VMs	Recovery Services vault	Vaulted	Same as VMs; point-in-time restore with log backups
Azure Files	Recovery Services vault	Snapshots plus vaulted backup	Vaulted backup generally available
Azure Blob (flat)	Backup vault	Operational (up to 360 days, source account only) and vaulted (up to 10 years)	Immutable vault and MUA for the vaulted tier; soft delete and versioning for the operational tier; container soft delete alongside
Azure Data Lake Storage (HNS)	Backup vault	Vaulted only; auto-protection up to 1,000 containers	Immutable vault, MUA
Managed disks	Backup vault	Incremental snapshots	Immutable vault
PostgreSQL flexible server	Backup vault	Vaulted, up to 10 years	Immutability
AKS, Elastic SAN	Backup vault	Per service	Immutability; WORM-backed storage available
Third-party backup images	Immutable Blob container	Not Azure Backup	Locked time-based retention; version-level WORM for object-lock semantics; Cool or Archive with minimums in mind
Azure Cosmos DB	Backup vault	Vaulted (preview)	Check current status

Vault limits that shape design

Limit	Value
Vaults per subscription	Up to 500 (Recovery Services or Backup)
Datasources per vault	Up to 2,000
Azure VMs per vault	Up to 1,000; 250 registrations per day
Backup storage redundancy	LRS, ZRS, or GRS; set before the first item is protected
Management plane	Resiliency in Azure (extends Azure Business Continuity Center)

Card 7. Ransomware resilience controls

Session 3

Assume the administrator is compromised. Each control answers a different attack.

Attack path	Control that stops it	Reversible?
Delete recovery points	Immutable vault	Enabled: yes. Locked: never
Reduce policy retention	Immutable vault and multi-user authorization	Via the guard owner
Disable soft delete	Enhanced soft delete always-on and MUA	AlwaysOn: never
Purge an immutable backup container	Locked time-based immutability on Blob	Never until expiry
Delete the storage account holding backup images	Locked time-based immutability on Blob (unlocked does not block account deletion)	Never until expiry
Act unnoticed	Built-in security alerts through Azure Monitor; activity log; Resource Guard log	Not applicable

Control	Detail
Immutable vault	Blocks operations that could lose recovery points; immutability based on policy retention or a specific duration; Enabled (reversible) or Locked (irreversible); WORM-backed storage generally available for Backup vaults in all public regions and for Recovery Services vaults in a growing set of regions, with the service enforcing locked immutability elsewhere and migrating automatically
Soft delete	On by default, 14 days at no cost; enhanced soft delete up to 180 days and always-on; soft-deleted data beyond the free window is billed
Multi-user authorization	Resource Guard in a separate subscription or tenant owned by a different team; the vault administrator alone cannot perform protected operations (delete protection, disable soft delete, reduce retention, disable immutability, remove MUA, and related)
Least privilege	Backup Reader, Backup Operator, Backup Contributor; narrowest role, narrowest scope, shortest time
Placement	Vaulted data lives in a Microsoft-managed subscription and tenant; vault shares the datasource's region and usually subscription (cross-subscription backup for Azure VMs and Azure Files in preview); put the guard, not the vault, elsewhere
Enforcement	Azure Policy requires backup, immutability, and soft delete; Resiliency in Azure reports status

Say it precisely to leadership

Locked immutable vault where WORM-backed storage has not yet arrived in your region; both block the same operations. Locked means locked: a locked vault can never be deleted.

Card 8. Redundancy option comparison

Session 4

Durability figures are Microsoft's design targets over a year. Redundancy copies mistakes too; pair it with soft delete, versioning, backup, and immutability.

Option	Copies and placement	Survives	Read from secondary	Durability	Notes
LRS	3 in one datacenter	Drive and rack failure	No	At least 11 nines	Lowest cost; no datacenter-level protection
ZRS	3 across zones, synchronous	Zone loss	No	At least 12 nines	Stays readable and writable if one zone is lost
GRS	LRS plus async copy in the paired region	Regional loss	No	16 nines	Recent writes can be lost on unplanned failover
GZRS	ZRS plus async copy in the paired region	Zone and regional loss	No	16 nines	Premium accounts cannot use geo options
RA-GRS	GRS with read access	Regional loss	Yes	16 nines	Application handles the secondary endpoint
RA-GZRS	GZRS with read access	Zone and regional loss	Yes	16 nines	

Account failover

Type	When	Data loss	Geo-redundancy after	Notes
Customer-managed planned	DR drills, proactive response	None; completes after synchronization	Retained; primary and secondary swap	Needs geo-redundancy configured and synced; requires the storage-preview CLI extension; account unavailable for writes during the swap; fail back with another planned failover
Customer-managed unplanned	Unexpected outage	Everything after last sync time	Lost; account becomes LRS until re-enabled (egress cost)	Check last sync time first; NFS 3.0 accounts support only unplanned
Conflicts				Object replication accounts cannot fail over; immutability policy changes after the last sync time do not carry over

Object replication versus geo-redundancy

	Geo-redundancy	Object replication
Scope	Whole account	Chosen containers and prefixes, up to 1,000 rules per policy
Destination	Fixed paired region	Up to two destination accounts in any region, subscription, or tenant (cross-tenant disabled by default for accounts created after December 15, 2023)
Prerequisites	None	Change feed on the source, versioning on both; GPv2 or premium block blob
Not supported	Premium accounts	HNS, NFS 3.0, SFTP accounts; append and page blobs; snapshots; blobs already in Archive; accounts using customer-managed failover
Lag	Asynchronous	Asynchronous; priority replication adds a 99 percent within 15 minutes objective on the same continent
Use	Durability	Independence: a separately administered copy, replicated while online then tiered to Archive

Card 9. RPO and RTO design method

Session 4

RPO: how much data you can afford to lose, in time. RTO: how long the workload can be down. Both are business decisions; the platform choice follows.

Step	Do	Output
1. Classify	Workloads by business impact	Four tiers
2. Target	Tiered RPO and RTO per class	Numbers signed by business owners
3. Map	Each tier to redundancy, replication, backup, and DR	Building blocks per tier
4. Price	Each tier with the cost model	Cost per tier
5. Document	The accepted risk and who accepted it	Risk register entries

Tier	Typical need	Azure building blocks
Mission critical	Minimal data loss, fast recovery	ZRS or GZRS; Site Recovery with recovery plans and multi-VM consistency; immutable backup with cross-region restore; Data Guard for Oracle
Business important	Hours of tolerance	GRS; backup with cross-region restore; rebuild rather than live replication
Research active	Must not lose data, can wait	Flat accounts: GRS or object replication, versioning, soft delete. HNS data lakes: GRS plus vaulted backup for Data Lake Storage
Research archive	Never lost, days to recover	Container-level WORM; second copy made while online (object replication for flat; AzCopy or Data Factory for HNS or archived data), then Archive at both ends; a copy outside Azure for independence

Site Recovery essentials

Item	Detail
Replication	Azure VMs between any two Azure regions; on-premises machines to Azure; data held in a Recovery Services vault
Recovery plans	Up to 100 protected instances; groups define boot order; manual actions and scripts; multi-VM consistency
Test failover	Isolated network; no effect on replication or production; clean up afterwards with notes
Recovery point choice	Latest processed (lowest RTO); latest app-consistent; latest (lowest RPO); latest multi-VM processed
Network mapping	Both directions; plan IPs, DNS, private endpoints in the target region; unmapped targets get an -asr network

Card 10. Oracle on Azure decision guide

Session 5

Consideration	Oracle AI Database@Azure (formerly Oracle Database@Azure)	Oracle on Azure virtual machines
Who runs the infrastructure	Oracle operations team through a connection to OCI	Your team
Platform	Exadata Database Service on Dedicated Infrastructure, Exadata Database Service on Exascale Infrastructure, Oracle Base Database Service, Oracle Autonomous AI Database; OCI infrastructure inside Azure datacenters in more than 30 regions	Memory-optimized VM sizes with Premium SSD, Premium SSD v2, or Ultra Disk; Azure NetApp Files as an NFS option
Data and protection services	Zero Data Loss Autonomous Recovery Service, OCI GoldenGate, Autonomous AI Lakehouse	RMAN, Azure Backup (application-consistent VM), Azure Files, Blob targets
Versions	19c and 26ai (23ai was renamed 26ai)	Any supported version you install
Purchase and licensing	Azure Marketplace private offer with a linked OCI tenancy; BYOL or license-included by service; counts toward Azure consumption commitments	Oracle's authorized cloud environment policy; confirm with the Oracle account team
Management	Infrastructure and VM clusters in the Azure portal; some container and pluggable database tasks in the OCI console; Azure networking, identity, monitoring	Full control: patching, Data Guard, backups
Best fit	Exadata-class performance with less operational burden	Maximum control and custom configuration

Immutable RMAN target pattern (Oracle on Azure VMs)

Element	Choice	Why
Mount	BlobFuse2 on a flat-namespace GPv2 account with managed identity (MSI) auth	Container-level WORM, versioning, GZRS, and object replication available; no keys on the VM
Not NFS 3.0 on Blob		Requires a hierarchical namespace account, LRS or ZRS only, and immutability policies are not supported on NFS 3.0-enabled accounts
Policy	Time-based retention equal to or longer than the RMAN recovery window; legal hold for audits; version-level WORM where object-lock semantics are needed	RMAN obsolete deletions succeed only after expiry
Geo	GZRS plus object replication to an immutable destination in a separate subscription, while pieces are online	Object replication cannot read archived pieces
Separation of duties	DBAs run RMAN; storage administrators own the target; Security owns the lock and the Resource Guard	No single role can create and destroy the copy
Validation	Weekly automated restore to an isolated environment; RMAN VALIDATE; control totals; measured RTO	Evidence, not hope

Other platforms

Platform	Primary protection	Long-term or immutable copy
SQL Server in Azure VMs	Azure Backup workload backup (full, differential, log; point-in-time restore)	Immutable Recovery Services vault
Azure SQL	Automatic backups, 1 to 35 days; log backups about every 10 minutes; Hyperscale uses storage snapshots	Long-term retention up to 10 years
Azure Database for PostgreSQL flexible server	Automatic backups, geo-restore	Backup vault with immutability, up to 10 years (generally available)
Azure Database for MySQL	Automatic backups, geo-restore	Azure Backup long-term retention is a paused preview; use native export

Card 11. FinOps and governance quick reference

Session 6

Lever	Rule	Watch out
Reserved capacity	One or three years; 100 TiB or 1 PiB units; Hot, Cool, and Archive in standard accounts; bound to one region, one tier, one redundancy; billing discount on capacity only	Not Cold, smart tier, or premium; buy at steady state
Tiering and lifecycle	Thresholds at or above tier minimums; target tags	Early deletion; small objects
Smart tier	Unpredictable access within Hot, Cool, Cold	ZRS GPv2 required; Hot access rates plus monitoring meter; excluded from reservations
Azure Files provisioned v2	Storage, IOPS, and throughput provisioned separately	Reservations cover SSD v1 and HDD pay-as-you-go, not v2
Redundancy right-sizing	Per data class from the resilience matrix	Account-wide
Backup retention tuning	Keep what policy requires	Retention length and vault redundancy are the biggest backup levers
Azure Advisor	Monthly input to the review	

Cost Management capability	Use
Cost analysis	Group by tag (ResearchUnit, DataClass, CostCenter); portal or Power BI
Tag inheritance	Subscription and resource group tags onto cost records (not onto resources)
Cost allocation rules	Split shared platform costs
Budgets and alerts	Actual and forecasted thresholds; anomaly, reservation utilization, scheduled alerts
Exports	Feed external tools; cost records lag usage by several hours to a day

Azure Policy effects

Effect	Does	Storage example
Audit	Reports non-compliance	Accounts without private endpoints
Deny	Blocks non-compliant requests	Public blob access enabled
Modify	Adds or changes properties	Add a missing tag
DeployIfNotExists	Deploys a related resource	Enable backup for a resource

Built-in storage policies used in the course (verify definition IDs by display name at deploy time): Secure transfer to storage accounts should be enabled; Storage accounts should have the specified minimum TLS version; Storage account public access should be disallowed; Storage accounts should prevent shared key access; Storage accounts should restrict network access; Require a tag on resources; Inherit a tag from the resource group if missing.

Data management as code

Element	Detail
Bicep modules	Storage account (SKU, HNS, network rules, minimum TLS, shared key), lifecycle policy child resource, container immutability and versioning, Backup vault with immutability, RBAC, tags; parameterize per data class
Deployment stacks	Manage the platform as one unit; actionOnUnmanage (delete or detach); deny settings; subscription scope includes the resource group because deleting a resource group bypasses deny assignments
Pipeline	Commit; lint and build; what-if preview; approval gate; deploy the stack with GitHub Actions or Azure Pipelines; workload identity federation, no stored secrets

Card 12. Azure CLI cheat sheet

All sessions

The commands used in the demos and labs. Extensions: dataprotection (Backup vaults), storage-preview (planned account failover).

Storage and protection

Task	Command
Create a baseline account	az storage account create -n NAME -g RG -l REGION --kind StorageV2 --sku Standard_ZRS --https-only true --min-tls-version TLS1_2 --allow-blob-public-access false --allow-shared-key-access false
Versioning, soft delete, change feed	az storage account blob-service-properties update -n NAME -g RG --enable-versioning true --enable-delete-retention true --delete-retention-days 14 --enable-container-delete-retention true --container-delete-retention-days 14 --enable-change-feed true
Last access tracking (not with WORM)	az storage account blob-service-properties update -n NAME -g RG --enable-last-access-tracking true
Lifecycle policy	az storage account management-policy create --account-name NAME -g RG --policy @policy.json
Upload with tags	az storage blob upload-batch --account-name NAME --auth-mode login -d CONTAINER -s DIR --tags ResearchUnit=PGRL RetentionClass=Permanent
Find blobs by tag	az storage blob filter --account-name NAME --auth-mode login --tag-filter "\"RetentionClass\"='Project'"
Archive and rehydrate	az storage blob set-tier ... --tier Archive; az storage blob set-tier ... --tier Cool --rehydrate-priority Standard
Container WORM create, lock	az storage container immutability-policy create --account-name NAME -c C -g RG --period 30; ... lock --if-match ETAG
ABAC assignment	az role assignment create --role "Storage Blob Data Reader" --assignee-object-id ID --assignee-principal-type User --scope SCOPE --condition "..." --condition-version "2.0"
Backup vault	az dataprotection backup-vault create -g RG --vault-name V -l REGION --type SystemAssigned --storage-setting "[{type:'LocallyRedundant',datastore-type:'VaultStore'}]" --immutability-state Unlocked --soft-delete-state On --retention-duration-in-days 14
Resource Guard and link	az dataprotection resource-guard create -g RGG --resource-guard-name G -l REGION; az dataprotection backup-vault resource-guard-mapping create -g RG --vault-name V -n DppResourceGuardProxy --resource-guard-id ID
Vaulted blob backup	initialize-backupconfig --datasource-type AzureBlob --storage-account-rg RG --storage-account-name NAME --container-list c1 c2; initialize --datasource-type AzureBlob --datasource-location REGION --policy-id P --datasource-id SAID --backup-configuration cfg.json; validate-for-backup; create; adhoc-backup --rule-name BackupDaily
Vaulted blob restore	az dataprotection recovery-point list ...; backup-instance restore initialize-for-item-recovery --datasource-type AzureBlob --restore-location REGION --source-datastore VaultStore --recovery-point-id RP --target-resource-id ALT_ID --container-list c1; validate-for-restore; restore trigger
Recovery Services vault posture	az backup vault create ... --immutability-state Unlocked; az backup vault backup-properties set -g RG -n V --soft-delete-feature-state AlwaysOn --soft-delete-duration 14 --backup-storage-redundancy GeoRedundant --cross-region-restore-flag True; az backup vault resource-guard-mapping update -g RG -n V --resource-guard-id ID

Resilience, deployment, and governance

Task	Command
Planned account failover	az extension add --name storage-preview; az storage account failover -n NAME -g RG --failover-type planned --yes
Last sync time	az storage account show -n NAME -g RG --expand geoReplicationStats --query geoReplicationStats
Deployment stack	az stack group create --name S --resource-group RG --template-file main.bicep --parameters main.bicepparam --action-on-unmanage deleteResources --deny-settings-mode denyDelete --yes
What-if	az deployment group what-if -g RG --template-file main.bicep --parameters main.bicepparam
Policy compliance scan	az policy state trigger-scan -g RG

Card 13. Certification pathway

Session 6 close

Role-based mapping; AZ-900 is assumed complete for every attendee.

Role	Path	What in this course maps
Storage infrastructure engineers	AZ-104 Azure Administrator Associate, then AZ-305 Azure Solutions Architect Expert	Sessions 1, 2, 4, 6: storage design, redundancy, resilience, governance
Data management specialists	AZ-104, then AZ-305; DP-203 Data Engineer Associate if data lakes grow	Sessions 1, 2, 6: tiering, archival, tags, cost allocation
Data protection and DR specialists	AZ-104, then AZ-305; SC-500 Cloud and AI Security Engineer Associate for the security controls	Sessions 3 and 4: vaults, immutability, MUA, Site Recovery
Oracle database infrastructure engineers	DP-300 Azure Database Administrator Associate; AZ-104 for the platform	Session 5: platform-native protection, Oracle on Azure patterns
Security-focused staff	SC-500 (replaced AZ-500 in August 2026)	Resource Guards, policy, identity, ABAC

Exam	Name	Level	Notes
AZ-104	Microsoft Certified: Azure Administrator Associate	Associate	Prerequisite for the Expert credential
AZ-305	Microsoft Certified: Azure Solutions Architect Expert	Expert	Requires AZ-104
DP-300	Microsoft Certified: Azure Database Administrator Associate	Associate	Azure SQL focus; relevant for platform-native protection
SC-500	Microsoft Certified: Cloud and AI Security Engineer Associate	Associate	Replaced AZ-500 in August 2026
DP-203	Microsoft Certified: Azure Data Engineer Associate	Associate	Optional, for data lake teams

Verify exam names, availability, and retirement dates on Microsoft Learn before registering; the certification catalog changes during the year.

Study approach the course supports

Each session's design exercise is a scenario question in miniature; the facilitator guides' model answers read like exam rationales. Pair the knowledge checks from the workbooks with Microsoft Learn's free practice assessments.